MCV4U CCT · Topic C · Alan and Jeff · Companion website with narrated videos and typed solutions.

Course Culminating Task · Topic C

Derivatives & Optimization

An interactive, curriculum-aligned calculus portal for MCV4U Grade 12. Learn differentiability, rectilinear motion, extreme values, and real-world optimization with worked problems and practice assessments.

Instructors: Alan & Jeff • Ontario Secondary School Curriculum

This is the PDF version of the website. Online, the site also has a live tangent-line simulator, graphs you can read by hovering, and embedded video lessons.

CURRICULUM SYLLABUS

Four Pillars of Topic C

Key mathematical topics covered in this culminating project according to MCV4U guidelines.

MODULE 01

Differentiability & Continuity

Foundations of calculus: formal limit definitions of derivatives, relationship between continuity and differentiability, and analysis of non-differentiable points including sharp corners, cusps, and vertical tangents.

- Left & Right Limits
- Corners, Cusps & Vertical Tangents
- Piecewise Functions

MODULE 02

Velocity, Acceleration & Motion

Rectilinear kinematics with calculus: analyzing position s(t), velocity v(t) = s′(t), and acceleration a(t) = v′(t). Rigorously determining intervals of accelerating vs. decelerating bodies via sign analysis.

- Position & Velocity
- Acceleration & Direction
- Speeding Up vs. Slowing Down

MODULE 03

Maxima, Minima & Extreme Values

Extreme Value Theorem (EVT) on closed intervals [a, b], identifying critical numbers (f′=0 or DNE), and applying the First and Second Derivative Tests to classify local and absolute extrema.

Critical Numbers | Closed Interval Method | 1st & 2nd Derivative Tests

MODULE 04

Multidisciplinary Optimization

Formulating and solving constrained optimization models with the 5-step framework across three required domains: open-top box design, business profit, and two simplified biology models (enzyme activity and drug concentration).

Industrial Design | Economics & Profit | Biological Sciences

VIDEO LESSONS

Instructional Videos

Two complete narrated lessons covering all four modules. Play either video below, then practise with the linked homework and assessments. Use the interactive animations to pause and explore each mathematical step.

Video 1 · Alan

Differentiability + Motion

Limit definition, continuity, one-sided derivatives, position, velocity, acceleration, direction and speed.

Alan · 15:30 · Recorded English narration

Open video in a separate tab Explore the interactive animation

▼ **Jump to a chapter**

0:00 · Opening problem 1:18 · From secant to tangent 4:00 · Microscope

5:32 · Cusps and vertical tangents 7:16 · Match values and slopes 8:53 · Bridge

9:20 · Position, velocity and acceleration 11:09 · Speeding up and slowing down

13:06 · Distance and displacement 14:26 · Review and next steps

Next: Homework 1 Homework 2 Assessment 1

Video 2 · Jeff

Extrema + Optimization

Critical values, the closed-interval method and worked examples from design, economics and biological science.

Jeff · 14:00 · Recorded English narration

Open video in a separate tab Explore the interactive animation

▼ Jump to a chapter

0:00 · Opening problem 1:06 · Critical numbers 2:21 · The closed-interval method

4:15 · Derivative tests 5:51 · Design: open-top box 7:55 · Economics: maximum profit

10:28 · Biology: drug concentration 12:01 · Yourturn 13:14 · Review and next steps

Next: Homework 3 Homework 4 Assessment 2

Mathematical clarification: The drug-model domain $[0, \infty)$ is closed but unbounded. The closed-and-bounded interval form of EVT does not apply. Instead, C' changes from positive to negative at $t = 2$, $C(0) = 0$, and $C(t) \to 0$ as $t \to \infty$; the absolute maximum is 2.5 mg/L at 2 hours.

FULL TEACHING MATERIAL

Four Guided Lessons

Each lesson moves from definitions to worked examples and graphs, then on to homework and assessment.

01

MODULE 01

Differentiability

Understand when a derivative exists and why continuity is necessary but not sufficient.

01. The derivative at a point

The derivative is the limit of the average rate of change. At an interior point a, a finite, two-sided limit must exist: $f'(a) = \lim_{h \to 0} [f(a+h) - f(a)] / h$. The left and right derivatives must agree. Differentiability at a implies continuity at a, but continuity alone does not guarantee differentiability.

WORKED EXAMPLE

For f(x) = x², calculate f′(3) from the definition.

1. [f(3+h) − f(3)]/h = [(3+h)² − 9]/h.
2. Expand to obtain (6h+h²)/h = 6+h for h ≠ 0.
3. Take the limit as h approaches 0: **f′(3) = 6**.

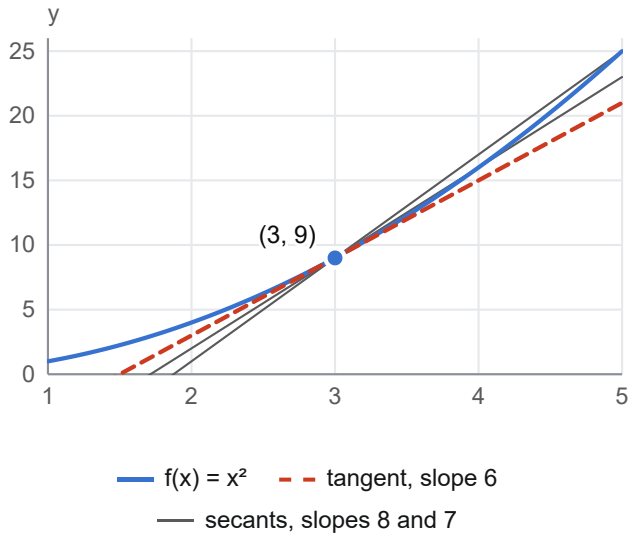


Figure 1.1. As h shrinks, the secant slopes 8 and 7 approach the tangent slope 6.

> Interpretation: at x = 3, the tangent slope is 6.

02. Continuous does not always mean differentiable

A corner has unequal one-sided slopes; a cusp or vertical tangent can have an undefined or infinite slope. A discontinuity also prevents differentiability.

WORKED EXAMPLE

Check $f(x) = |x - 2|$ at $x = 2$.

1. The function is continuous: $f(2) = 0$ and both one-sided limits equal 0.
2. For $x < 2$, $f(x) = 2 - x$, so the left derivative is -1.
3. For $x > 2$, $f(x) = x - 2$, so the right derivative is $+1$.
4. The one-sided derivatives disagree; therefore **$f'(2)$ does not exist**.

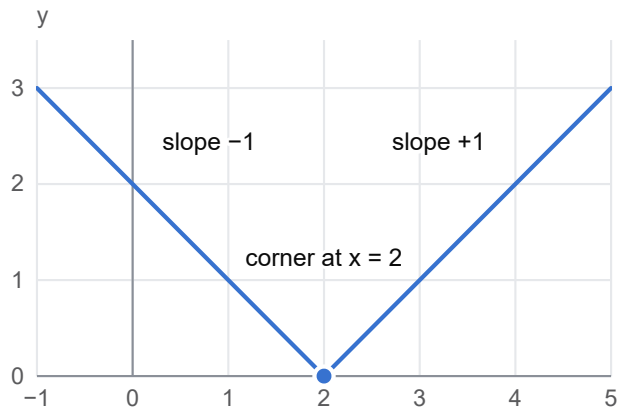


Figure 1.2. A corner: the slopes on each side of $x = 2$ are different.

> A continuous corner is a counterexample to the converse statement.

03. Cusps and vertical tangents

At a cusp and at a vertical tangent the curve is continuous, but the difference quotient grows without bound as $h \to 0$, so no finite derivative exists. At a cusp the one-sided slopes head to opposite infinities. At a vertical tangent they head to the same infinity.

WORKED EXAMPLE

Test $f(x) = x^{2/3}$ and $g(x) = x^{1/3}$ at $x = 0$.

1. For f: $[f(h) - f(0)]/h = h^{2/3}/h = 1/h^{1/3}$.
2. As $h \to 0^+$ this tends to $+\infty$, and as $h \to 0^-$ it tends to $-\infty$. The slopes disagree, so f has a **cusp** and $f'(0)$ does not exist.
3. For g: $[g(h) - g(0)]/h = h^{1/3}/h = 1/h^{2/3}$.
4. As $h \to 0$ from either side this tends to $+\infty$. The tangent is the vertical line $x = 0$, so g has a **vertical tangent** and $g'(0)$ does not exist.

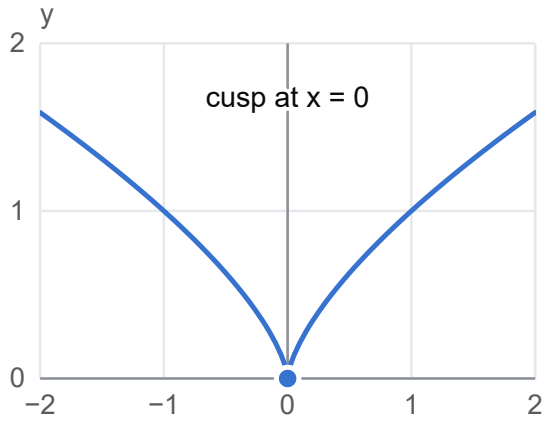


Figure 1.3. Cusp: $f(x) = x^{2/3}$.

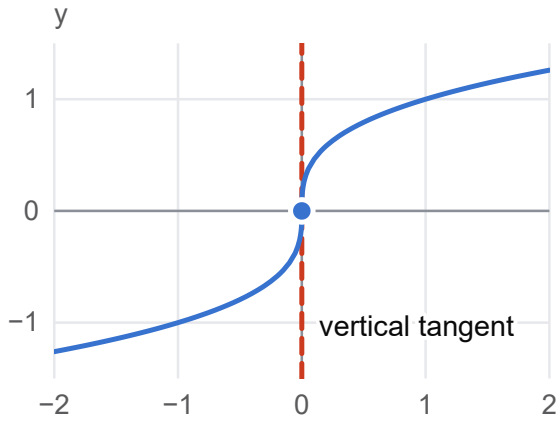


Figure 1.4. Vertical tangent: $g(x) = x^{1/3}$.

> The derivative formulas agree: $f'(x) = (2/3)x^{-1/3}$ and $g'(x) = (1/3)x^{-2/3}$ are both undefined at $x = 0$.

04. Piecewise functions

At a joining point, first match function values (continuity), then match one-sided derivatives (smoothness). Checking slopes alone is not enough.

WORKED EXAMPLE

Let $f(x) = x^2$ for $x \le 1$ and $f(x) = mx+b$ for $x > 1$. Find m and b for differentiability at 1.

1. Continuity requires $m+b = 1$, because the left value equals 1.
2. The left derivative at 1 is 2; the right derivative is m. Set $m = 2$.
3. Then $2+b = 1$, so **$b = -1$**.
4. Both continuity and one-sided derivative conditions hold, so f is differentiable at 1.

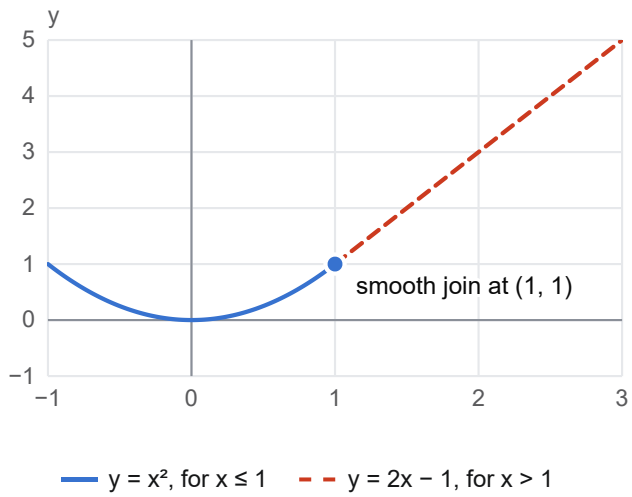


Figure 1.5. With m = 2 and b = −1, the two pieces share a value and a slope at x = 1.

> Always test continuity before differentiability.

02

MODULE 02

Velocity & Acceleration

Connect derivatives to one-dimensional motion, direction, speed, and acceleration.

01. Position, velocity, acceleration

For a position function s(t), velocity is `v(t) = s′(t)` and acceleration is `a(t) = v′(t) = s″(t)`. Position is measured in metres (m), velocity in m/s and acceleration in m/s². Speed is |v(t)|, not v(t).

WORKED EXAMPLE

A particle has $s(t)=t^3-6t^2+9t$ for $0 \le t \le 5$.

1. Differentiate: **$v(t)=3t^2-12t+9=3(t-1)(t-3)$**.
2. Differentiate again: **$a(t)=6t-12=6(t-2)$**.
3. The particle is at rest when v=0, so t = 1 and t = 3 seconds.

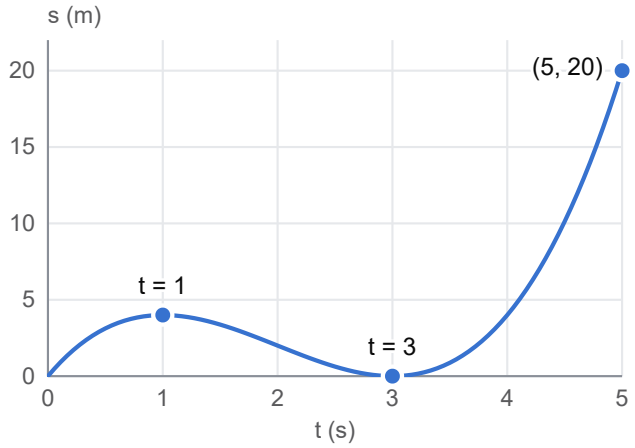


Figure 2.1. The particle stops and turns around at t = 1 and t = 3.

> Positive velocity means motion to the right; negative velocity means motion to the left.

02. Speeding up versus slowing down

Speed increases when velocity and acceleration have the same sign, $v \cdot a > 0$; it decreases when signs differ, $v \cdot a < 0$. At zeros of v or a, discuss nearby intervals rather than declaring the object faster or slower at that instant.

WORKED EXAMPLE

Classify the particle above on (0,5).

1. Split the time domain at t=1, 2 and 3.
2. On (0,1): v>0 and a<0 → slowing down.
3. On (1,2): v<0 and a<0 → speeding up.
4. On (2,3): v<0 and a>0 → slowing down.
5. On (3,5): v>0 and a>0 → speeding up.

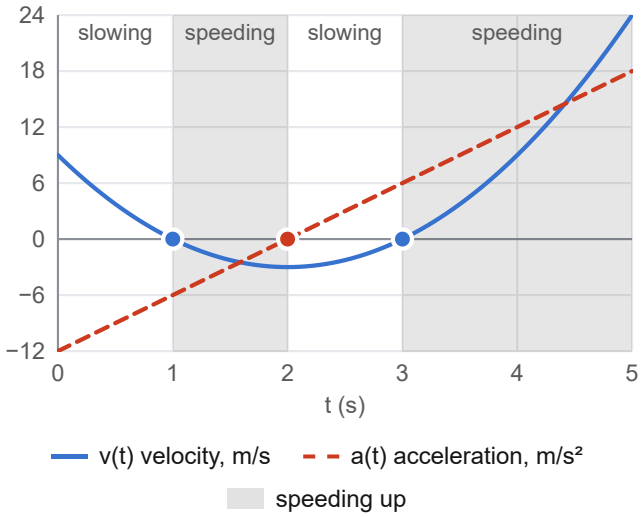


Figure 2.2. Speed increases where v and a have the same sign (shaded).

> Acceleration does not always mean “speeding up”. Its direction matters relative to velocity.

03. Displacement versus total distance

Displacement is final position minus initial position. Total distance travelled adds the absolute changes in position between turning times when velocity is zero.

WORKED EXAMPLE

Find the displacement and total distance for the same particle on [0,5].

1. Position values: s(0)=0; s(1)=4; s(3)=0; s(5)=20.
2. Displacement = s(5)−s(0) = **20 m**.
3. Total distance = |4−0| + |0−4| + |20−0| = **28 m**.

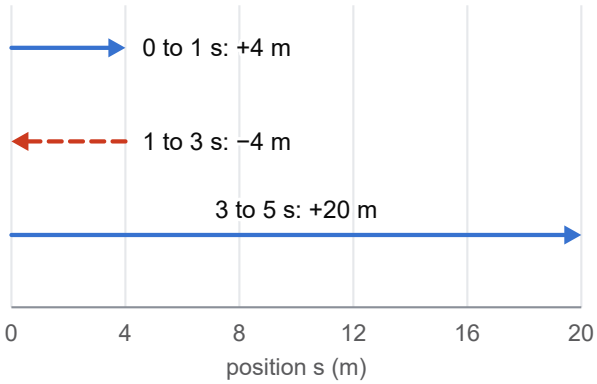


Figure 2.3. Distance adds every leg: 4 + 4 + 20 = 28 m.
Displacement only compares the start and the end: 20 m.

> Distance is nonnegative and never smaller than the magnitude of displacement.

03

MODULE 03

Maxima, Minima & Extreme Values

Find and justify local and absolute extremes using derivative tests and endpoints.

01. Critical numbers and the EVT

A critical number is an interior domain value where $f'(x)=0$ or $f'(x)$ does not exist. The Extreme Value Theorem says that a function continuous on the closed, bounded interval $[a,b]$ attains an absolute maximum and an absolute minimum there. Candidates include critical numbers and endpoints.

WORKED EXAMPLE

Find absolute extrema of $f(x)=x^3-3x$ on $[-2,2]$.

1. Compute $f'(x)=3x^2-3=3(x-1)(x+1)$.
2. Critical numbers inside $[-2,2]$: $x=-1$ and $x=1$.
3. Evaluate: $f(-2)=-2$, $f(-1)=2$, $f(1)=-2$, $f(2)=2$.
4. Absolute maximum: **2 at $x=-1$ and $x=2$**. Absolute minimum: **-2 at $x=-2$ and $x=1$**.

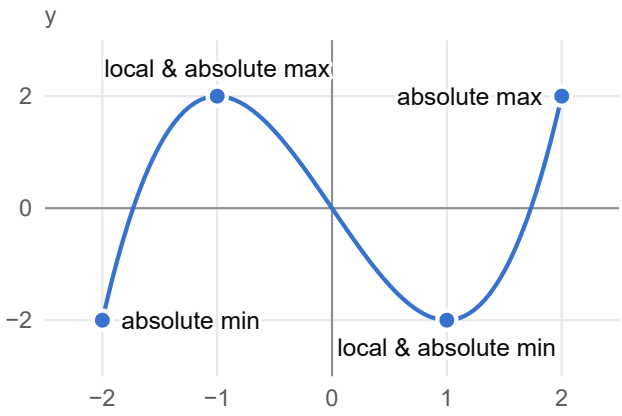


Figure 3.1. The endpoints tie with the turning points for the absolute extrema.

> Do not forget the endpoints in a closed-interval problem.

02. First and second derivative tests

For the first derivative test, inspect the sign of f' before and after a critical number: + to − means a local maximum; − to + means a local minimum. If $f'(c)=0$, a positive $f''(c)$ means local minimum; negative $f''(c)$ means local maximum. If $f''(c)=0$, the test is inconclusive.

WORKED EXAMPLE

Classify the interior critical numbers of $f(x)=x^3-3x$ with both tests.

1. **First derivative test.** $f'(x)=3(x-1)(x+1)$, so check the sign of f' on each side of $x=-1$ and $x=1$.

Interval	$x < -1$	$-1 < x < 1$	$x > 1$
Test value	$f'(-2) = 9$	$f'(0) = -3$	$f'(2) = 9$
Sign of f'	+	−	+
f is	increasing	decreasing	increasing

2. f' changes from + to − at $x=-1$, so there is a local maximum. It changes from − to + at $x=1$, so there is a local minimum.
3. **Second derivative test.** $f''(x)=6x$. At $x=-1$, $f''(-1)=-6<0$, so there is a local maximum.
4. At $x=1$, $f''(1)=6>0$, so there is a local minimum. Both tests agree.

> A stationary point need not be a maximum or minimum: $f(x)=x^3$ at $x=0$ is a counterexample.

03. A reliable optimization checklist

Use: (1) define the domain, (2) obtain a one-variable function, (3) differentiate, (4) identify critical points and endpoints, (5) justify and interpret the result. A derivative of zero identifies a candidate, not proof of a maximum.

WORKED EXAMPLE

Consider $g(x)=x^2$ on $[-2,3]$.

1. $g'(x)=2x=0$ gives $x=0$.
2. Evaluate: $g(-2)=4$, $g(0)=0$, $g(3)=9$.
3. The absolute minimum is 0 at $x=0$; the absolute maximum is 9 at $x=3$.

> Comparing candidate values confirms absolute extrema on a continuous closed interval.

04

MODULE 04

Optimization: Three Real-World Fields

Model a realistic objective, impose constraints, solve and explain the meaning of the optimum.

01. Design: open-top box

Required field: design. A 20 cm × 20 cm square sheet has identical corner squares removed, then folds into an open-top box. The cut size x must satisfy $0 < x < 10$. We maximize volume, not surface area.

WORKED EXAMPLE

Find the corner-cut size with greatest volume.

1. Base side length = $20-2x$; height = x. Hence $V(x)=x(20-2x)^2$.
2. $V'(x)=(20-2x)(20-6x)$.
3. Set $V'=0$: $x=10$ (boundary/degenerate box) or $x=10/3$ (interior).
4. V approaches 0 at each boundary; volume is positive inside. Thus **$x=10/3$ cm ≈ 3.33 cm** maximizes volume.
5. $V_{max} = (10/3)(40/3)^2 =$ **$16000/27$ cm³ ≈ 592.59 cm³**.

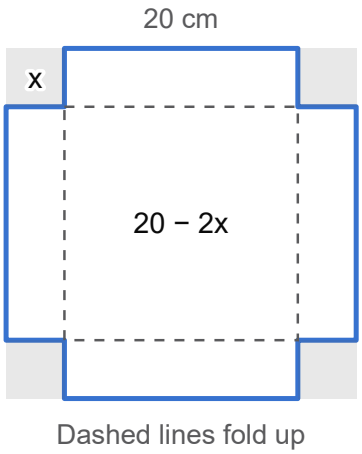


Figure 4.1. Cut four corners, then fold up the sides.

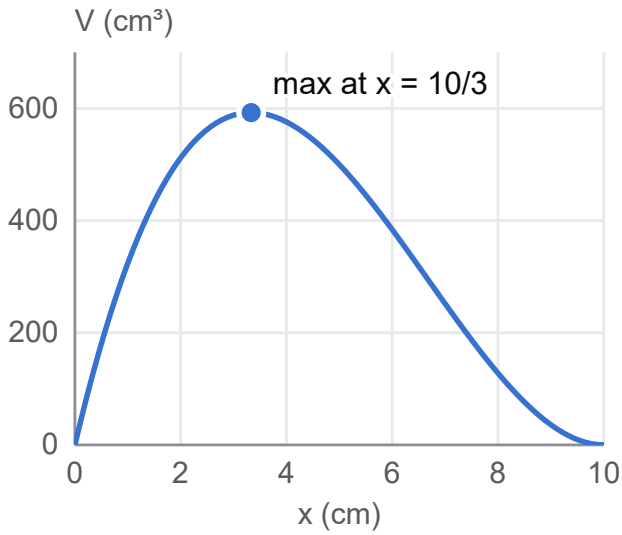


Figure 4.2. Volume is 0 at both ends and peaks at x = 10/3.

> The domain and boundary behavior are part of the justification. Here V' is positive for $0 < x < 10/3$ and negative for $10/3 < x < 10$, so this is the unique absolute maximum.

02. Economics: maximum profit

Required field: economics. In a simplified demand model, price is $p(q)=100-2q$ dollars per item, fixed and variable costs are $C(q)=200+20q$ dollars. Demand requires $0 \le q \le 50$. Profit is revenue minus cost.

WORKED EXAMPLE

Choose production q that maximizes profit.

1. $R(q)=q\cdot p(q)=100q-2q^2$.
2. $P(q)=R(q)-C(q)=80q-2q^2-200$.
3. $P'(q)=80-4q$; solve $P'=0 \to q=20$. Equivalently, marginal revenue $R'(q)=100-4q$ equals marginal cost $C'(q)=20$ at $q=20$.
4. $P''(q)=-4<0$, so the profit curve is concave down on $[0,50]$ and $q=20$ gives the absolute maximum.
5. Maximum profit: **$P(20)=\$600$**, at **20 items**.

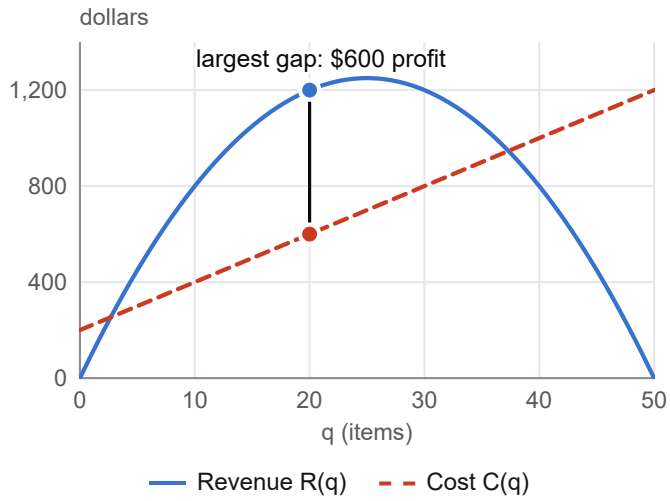


Figure 4.3. Profit is the gap between R and C. At q = 20 the two curves have equal slopes (MR = MC).

> MR=MC is a necessary interior stationary condition, not automatic proof of maximum profit.

03. Biological science: enzyme activity

Required field: biological science. Model enzyme activity in a hypothetical classroom experiment as $A(T)=10-0.02(T-35)^2$, where T is temperature (°C), $20 \le T \le 50$ and A is an activity index (arbitrary units). This is a pedagogical model, not measured biological data.

WORKED EXAMPLE

At what temperature is predicted activity greatest?

1. $A'(T)=-0.04(T-35)$.
2. Set $A'=0$ to obtain $T=35°C$.
3. $A''(T)=-0.04<0$ and 35 lies in the allowed interval.
4. The model predicts maximum **$A(35)=10$ activity units at 35°C**.

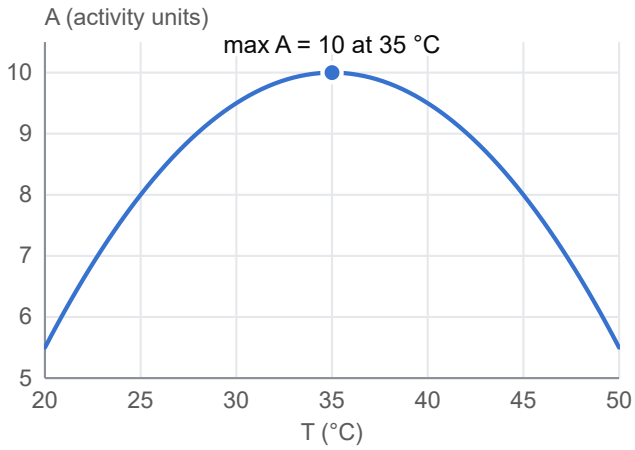


Figure 4.4. The model peaks inside the allowed temperature range.

> Model assumptions and validity ranges matter when interpreting applications.

04. Biological science: drug concentration

A second biology example. After a dose, a drug is absorbed into the bloodstream and then cleared. A simplified model is $C(t) = 10t/(t^2+4)$, where C is the concentration (mg/L) and $t \geq 0$ is the time in hours after the dose. The numbers are illustrative, not medical data.

WORKED EXAMPLE

When does the concentration peak, and what is the peak?

1. Quotient rule: $C'(t) = [10(t^2+4) - 10t(2t)]/(t^2+4)^2 = 10(4-t^2)/(t^2+4)^2$.
2. Set $C'=0$: $4-t^2=0$, so $t=2$. The value $t=-2$ is outside the domain.
3. $C'>0$ for $0 \leq t<2$ and $C'<0$ for $t>2$, so by the first derivative test $t=2$ is a maximum.
4. Peak: $C(2) = 20/8 =$ **2.5 mg/L, 2 hours after the dose**.

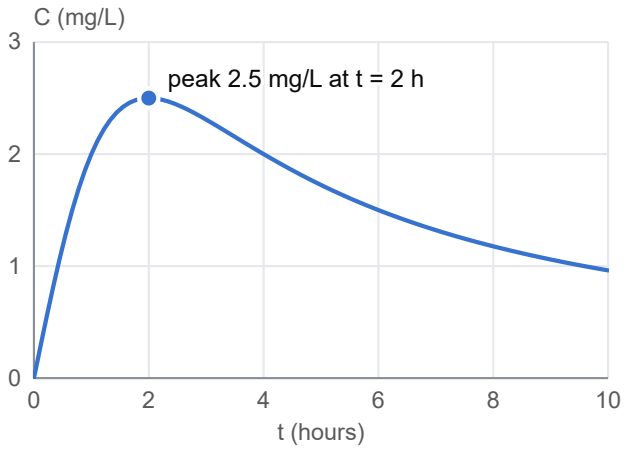


Figure 4.5. Concentration rises quickly, peaks at $t = 2$, then slowly falls.

> C rises, peaks once, then falls toward 0, so this local maximum is also the absolute maximum on $t \geq 0$.

PRACTICE

Four Homework Sets

18 worked problems spanning every required subtopic. Try each before revealing the solution.

HW 1 · Differentiability

5 questions · solutions provided

1.1 Determine whether $f(x)=|x+1|$ is differentiable at $x=-1$. Explain with one-sided derivatives.

Solution

1. For $x<-1$, $f(x)=-x-1$, so the left derivative is -1.
2. For $x>-1$, $f(x)=x+1$, so the right derivative is $+1$.
3. Both pieces meet continuously at $f(-1)=0$, but slopes differ. Therefore the function is not differentiable at $x=-1$.

1.2 Find a and b such that $f(x)=x^2+1$ for $x\le 2$ and $f(x)=ax+b$ for $x>2$ is differentiable at $x=2$.

Solution

1. Continuity: $2a+b = f(2)=2^2+1=5$.
2. Left derivative: $d(x^2+1)/dx$ at $x=2$ equals 4. Right derivative equals a.
3. Set $a=4$, then $8+b=5$ gives $b=-3$.

1.3 Is $f(x)=3x-1$ for $x<1$ and $f(x)=x^2+1$ for $x\ge 1$ differentiable at $x=1$?

Solution

1. Left limit $3(1)-1=2$; right limit and function value $1^2+1=2$. Hence continuous.
2. Left derivative is 3. Right derivative at 1 is $2(1)=2$.
3. Different one-sided derivatives mean $f'(1)$ does not exist.

1.4 Show from the limit definition that $f(x)=x^2$ has derivative $2x$ at $x=a$.

Solution

1. $[f(a+h)-f(a)]/h = [(a+h)^2-a^2]/h$.
2. Simplify to $[2ah+h^2]/h = 2a+h$ for $h \neq 0$.
3. As $h\to 0$ the limit is $2a$, so $f'(a)=2a$.

1.5 Use the limit definition to show that $f(x)=x^{4/5}$ is not differentiable at $x=0$. Does the graph have a cusp or a vertical tangent there?

Solution

1. f is continuous at $x=0$, because $h^{4/5}\to 0=f(0)$ as $h\to 0$.
2. The difference quotient is $[f(h)-f(0)]/h = h^{4/5}/h = 1/h^{1/5}$.
3. As $h\to 0^{+}$ this tends to $+\infty$, and as $h\to 0^{-}$ it tends to $-\infty$.
4. The one-sided slopes head to opposite infinities, so $f'(0)$ does not exist and the graph has a cusp at $x=0$.

HW 2 · Velocity & Acceleration

4 questions · solutions provided

2.1 For $s(t)=2t^3-9t^2+12t$ (metres), find $v(t)$, $a(t)$, $v(3)$, and $a(3)$.

Solution

1. $v(t)=s'(t)=6t^2-18t+12$.
2. $a(t)=v'(t)=12t-18$.
3. $v(3)=54-54+12=12$ m/s; $a(3)=36-18=18$ m/s².

2.2 For $v(t)=t^2-5t+6$ on $0\le t\le 4$, find when the particle is at rest and the intervals where it is speeding up.

Solution

1. Factor $v(t)=(t-2)(t-3)$, so the particle is at rest at $t=2$ and $t=3$.
2. $a(t)=v'(t)=2t-5$, so $a=0$ at $t=2.5$.
3. On $(2,2.5)$, $v<0$ and $a<0$. On $(3,4)$, $v>0$ and $a>0$. The signs match, so these are the speeding-up intervals.
4. On $(0,2)$ and $(2.5,3)$, v and a have opposite signs, so the particle is slowing down there.

2.3 For $s(t)=t^2-4t$ on $[0,5]$, calculate displacement and total distance.

Solution

1. $v(t)=2t-4=0$ at $t=2$.
2. $s(0)=0$; $s(2)=-4$; $s(5)=5$.
3. Displacement $=5-0=5$ m; total distance $=|-4-0|+|5-(-4)|=4+9=13$ m.

2.4 Explain why an object with $v=-4$ m/s and $a=-2$ m/s² is speeding up at that instant.

Solution

1. Velocity is negative, so motion is in the negative direction.
2. Acceleration is also negative, so it increases the magnitude of negative velocity.
3. Because $v \cdot a = 8 > 0$, its speed is increasing.

HW 3 · Maxima & Minima

5 questions · solutions provided

3.1 Find absolute extrema of $f(x)=x^2-6x+8$ on $[0,5]$.

Solution

1. $f'(x)=2x-6=0$ at $x=3$.
2. $f(0)=8$; $f(3)=9-18+8=-1$; $f(5)=25-30+8=3$.
3. Absolute max $=8$ at $x=0$; absolute min $=-1$ at $x=3$.

3.2 Find the absolute extrema of $f(x)=2x^3-3x^2-12x+1$ on $[-2,3]$.

Solution

1. $f'(x)=6x^2-6x-12=6(x-2)(x+1)$, so the critical numbers are $x=-1$ and $x=2$. Both are inside the interval.
2. Evaluate: $f(-2)=-3$; $f(-1)=8$; $f(2)=-19$; $f(3)=-8$.
3. Absolute maximum $=8$ at $x=-1$. Absolute minimum $=-19$ at $x=2$.

3.3 Classify the stationary points of $f(x)=x^4-4x^2$ on $\mathbb{R}$.

Solution

1. $f'(x)=4x^3-8x=4x(x^2-2)$. Critical points $x=0, \pm\sqrt{2}$.
2. $f''(x)=12x^2-8$. At 0 it is $-8<0$, so 0 is a local maximum with $f(0)=0$.
3. At $\pm\sqrt{2}$ it equals $16>0$, so these are local minima, each with value $(\sqrt{2})^4-4(\sqrt{2})^2=4-8=-4$.

3.4 Does $f(x)=x$ for $0 \le x < 1$ and $f(1)=0$ have an absolute maximum on $[0,1]$? Explain EVT.

Solution

1. For any $x<1$, there is another y between x and 1 with $f(y)=y>x$.
2. At $x=1$ the function equals 0. Therefore it never reaches its supremum 1 and has no maximum.
3. The function is discontinuous at 1, so EVT does not apply.

3.5 Use the first derivative test to classify the critical numbers of $f(x)=x^4-4x^3$. Why would the second derivative test not settle $x=0$?

Solution

1. $f'(x)=4x^3-12x^2=4x^2(x-3)$, so the critical numbers are $x=0$ and $x=3$.
2. Test the sign of f': $f'(-1)=-16<0$; $f'(1)=-8<0$; $f'(4)=64>0$.
3. At $x=0$, f' stays negative on both sides, so $x=0$ is neither a maximum nor a minimum.
4. At $x=3$, f' changes from $-$ to $+$, so $x=3$ is a local minimum, with $f(3)=81-108=-27$.
5. $f''(x)=12x^2-24x$ gives $f''(0)=0$, so the second derivative test is inconclusive at $x=0$. The first derivative test still gives an answer.

HW 4 · Optimization Applications

4 questions · solutions provided

4.1 A 24 cm × 9 cm rectangular sheet becomes an open-top box by cutting a square of side x from each corner. Find the x that maximizes the volume, and state that volume.

Solution

1. The base is $(24-2x)$ by $(9-2x)$ and the height is x, so $V(x)=x(24-2x)(9-2x)=4x^3-66x^2+216x$, with $0<x<4.5$.
2. $V'(x)=12x^2-132x+216=12(x-2)(x-9)$. The value $x=9$ is outside the domain, so $x=2$ is the only candidate.
3. V' is positive for $0<x<2$ and negative for $2<x<4.5$; V approaches 0 at both boundaries. Thus $x=2$ gives the absolute maximum. Also $V''(2)=-84<0$ confirms a local maximum.
4. $V(2)=2(20)(5)=200$ cm³, reached by cutting 2 cm squares.

4.2 A product has price $p(q)=80-q$ and cost $C(q)=100+20q$, $0\le q\le 80$. Maximize profit.

Solution

1. Revenue: $R(q)=q\cdot p(q)=80q-q^2$.
2. Profit: $P(q)=R(q)-C(q)=60q-q^2-100$.
3. $P'(q)=60-2q=0$ gives $q=30$, which is inside $[0,80]$.
4. $P''(q)=-2<0$, so $q=30$ is a maximum. The endpoints are lower: $P(0)=-100$ and $P(80)=-1700$.
5. Maximum profit: $P(30)=1800-900-100=\$800$, at 30 units.

4.3 A simplified model for the energy a migrating bird uses per kilometre at speed v (m/s) is $E(v)=0.05v^2+100/v$, for $2\le v\le 25$. Find the speed that uses the least energy, and explain why flying very slowly is not efficient.

Solution

1. $E'(v)=0.1v-100/v^2$.
2. Set $E'=0$: $0.1v^3=100$, so $v^3=1000$ and $v=10$ m/s.
3. $E''(v)=0.1+200/v^3>0$, so $v=10$ is a minimum.
4. Compare the endpoints: $E(2)=50.2$; $E(10)=15$; $E(25)=35.25$. The least energy is 15 units per km at 10 m/s.
5. At low speed the $100/v$ term becomes large: the trip takes longer, so each kilometre costs more energy. This is a classroom model with arbitrary units.

4.4 A garden against a straight wall requires 80 m of fencing for only three sides. Find maximum area.

Solution

1. Let x be each perpendicular side and y the parallel fenced side: $2x+y=80$.
2. Area $A=xy=x(80-2x)$, $0\le x\le 40$.
3. $A'=80-4x=0 \rightarrow x=20$; then $y=40$. $A''=-4<0$.
4. The maximum area is $20\cdot 40=800$ m².

ASSESSMENT CENTRE

Three Assessments + Final Test

Every assessment covers all four KATC categories and includes full step-by-step solutions. Try each question first, then reveal the solution.

K Knowledge & Understanding | A Application | T Thinking | C Communication

Assessment 1 · Differentiability & Motion

5 questions · 18 marks · K3 / A8 / T4 / C3

K / 3 **Q1.** State the relationship between differentiability and continuity. Give a counterexample to the converse.

Solution

1. If f is differentiable at x=a, then f is continuous at x=a.
2. The converse is false: a continuous function does not have to be differentiable.
3. Counterexample: f(x)=|x| is continuous at x=0, but its one-sided derivatives are −1 and +1, so f′(0) does not exist.

A / 4 **Q2.** Let $f(x)=x^2-3x$ for $x\le 2$ and $f(x)=mx+b$ for $x>2$. Find m and b so that f is differentiable at x=2.

Solution

1. Continuity: $f(2)=4-6=-2$, so $2m+b=-2$.
2. Left derivative: $2x-3$ at x=2 equals 1. Right derivative: m. Therefore m=1.
3. Substitute: $2(1)+b=-2$, so b=−4.
4. With m=1 and b=−4, the values and the slopes match, so f is differentiable at x=2.

A / 4 **Q3.** A particle has position $s(t)=t^3-3t^2$ (s in metres, t in seconds, $t\ge 0$). Find its velocity and acceleration, then describe its motion at t=1.5.

Solution

1. $v(t)=3t^2-6t=3t(t-2)$, in m/s.
2. $a(t)=6t-6$, in m/s².
3. $v(1.5)=3(1.5)(-0.5)=-2.25$ m/s and $a(1.5)=3$ m/s².
4. The particle is moving in the negative direction. v and a have opposite signs, so it is slowing down.

T / 4 **Q4.** Generalize Q3. For $s(t)=t^3-kt^2$ with $k>0$ and $t\geq 0$, find in terms of k the interval on which the particle slows down. Then find the value of k that makes this interval (2,4).

Solution

1. $v(t)=3t^2-2kt=t(3t-2k)$, so $v=0$ at $t=0$ and $t=2k/3$.
2. $a(t)=6t-2k$, so $a=0$ at $t=k/3$.
3. On $(0,k/3)$, $v<0$ and $a<0$: speeding up. On $(k/3,2k/3)$, $v<0$ and $a>0$: slowing down. For $t>2k/3$, $v>0$ and $a>0$: speeding up.
4. The particle slows down on $(k/3,2k/3)$. Setting $k/3=2$ gives $k=6$, and then $2k/3=4$ as required.

C / 3 **Q5.** A classmate writes: “In Q3 the acceleration at $t=1.5$ is positive, so the particle must be speeding up.” Explain the error in words.

Solution

1. Positive acceleration means the velocity is increasing, which is not the same as the speed increasing.
2. At $t=1.5$ the velocity is -2.25 m/s, so the particle moves in the negative direction while the acceleration acts in the positive direction. The velocity becomes less negative, so the speed $|v|$ gets smaller.
3. The correct test compares signs: $v\cdot a=(-2.25)(3)<0$, so the particle is slowing down.

Assessment 2 · Extreme Values

4 questions · 14 marks · K2 / A5 / T4 / C3

K / 2 **Q1.** Define a critical number of f on its domain.

Solution

1. First check that $f(c)$ is defined and c is in the interior of the domain.
2. Then c is a critical number if $f'(c)=0$ or $f'(c)$ does not exist.
3. Endpoints are checked separately when finding absolute extrema; a critical number is only a candidate, not proof of an extremum.

A / 5 **Q2.** Find the absolute extrema of $f(x)=x^2-4x+1$ on $[-1,4]$.

Solution

1. $f'(x)=2x-4=0$ at $x=2$, which is inside $[-1,4]$.
2. Evaluate the candidates: $f(-1)=6$; $f(2)=-3$; $f(4)=1$.
3. f is continuous on a closed interval, so the largest and smallest values are the absolute extrema: maximum 6 at $x=-1$, minimum -3 at $x=2$.

T / 4 **Q3.** Find a and b so that $f(x)=x^3+ax^2+bx$ has a local maximum at $x=-1$ and a local minimum at $x=3$. Verify your answer.

Solution

1. $f'(x)=3x^2+2ax+b$, and both $x=-1$ and $x=3$ must be zeros of f'.
2. So $f'(x)=3(x+1)(x-3)=3x^2-6x-9$.
3. Compare coefficients: $2a=-6$ gives $a=-3$, and $b=-9$.
4. Verify with $f''(x)=6x-6$: $f''(-1)=-12<0$ gives a local maximum, and $f''(3)=12>0$ gives a local minimum.

C / 3 **Q4.** State the Extreme Value Theorem and explain why continuity and a closed interval both matter.

Solution

1. EVT: if f is continuous on a closed, bounded interval [a,b], then f attains an absolute maximum and an absolute minimum on [a,b].
2. Continuity matters because a jump can skip the highest value: $f(x)=x$ for $0\le x<1$ with $f(1)=0$ has no maximum on [0,1].
3. A closed interval matters because an open one can leave out the extreme value: $f(x)=x$ on (0,1) gets close to 1 but never reaches it.

Assessment 3 · Applied Optimization

5 questions · 20 marks · K3 / A10 / T4 / C3

K / 3 **Q1.** In an optimization problem, define (a) the objective function and (b) a constraint. Then state what must be checked after solving $f'(x)=0$.

Solution

1. (a) The objective function is the quantity being maximized or minimized.
2. (b) A constraint is a condition that limits the variables. It reduces the problem to one variable and sets the domain.
3. After solving $f'(x)=0$, check any points where f' is undefined, justify the candidates with a derivative test, and compare included endpoints or relevant boundary limits.

A / 5 **Q2. Design.** A closed cylindrical can must hold 500 cm³. Find the radius and height that use the least material.

Solution

1. Constraint: $\pi r^2 h = 500$, so $h = 500/(\pi r^2)$, with $r > 0$.
2. Objective: $S(r) = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 1000/r$.
3. $S'(r) = 4\pi r - 1000/r^2 = 0$ gives $r^3 = 250/\pi$, so $r \approx 4.30$ cm.
4. $S''(r) = 4\pi + 2000/r^3 > 0$ for $r > 0$, so this is a minimum.
5. $h = 500/(\pi r^2) \approx 8.60$ cm, which equals $2r$. Since $S'' > 0$ for every $r > 0$ and $S \to \infty$ as $r \to 0^+$ or $r \to \infty$, this is the absolute minimum. The least material is about 348.7 cm².

A / 5 **Q3. Economics.** A theatre sells 300 tickets at \$20 each. A survey suggests that each \$1 price reduction sells 25 more tickets. What price maximizes revenue?

Solution

1. Let x be the number of \$1 reductions, $0 \le x \le 20$. Price is $20 - x$ and tickets sold are $300 + 25x$.
2. $R(x) = (20 - x)(300 + 25x) = 6000 + 200x - 25x^2$.
3. $R'(x) = 200 - 50x = 0$ gives $x = 4$.
4. $R''(x) = -50 < 0$, so $x = 4$ is a maximum. The endpoints are lower: $R(0) = 6000$ and $R(20) = 0$.
5. Charge \$16 to sell 400 tickets, for maximum revenue $R(4) = \$6400$.

T / 4 **Q4. Biology.** During a cough the windpipe narrows. A classic simplified model for the air speed is $v(r) = k(R - r)r^2$, where R is the normal radius, r is the narrowed radius with $R/2 \le r \le R$, and $k > 0$ is a constant. Find the r that maximizes air speed and explain what it means.

Solution

1. $v(r) = k(Rr^2 - r^3)$, so $v'(r) = k(2Rr - 3r^2) = kr(2R - 3r)$.
2. $v' = 0$ at $r = 0$, which is outside the domain, and at $r = 2R/3$.
3. Compare the candidates: $v(R/2) = kR^3/8 = 0.125kR^3$; $v(2R/3) = 4kR^3/27 \approx 0.148kR^3$; $v(R) = 0$.
4. The maximum is at $r = 2R/3$. In this model, air moves fastest when the windpipe narrows to about two-thirds of its normal radius, which helps to clear it.

C / 3 **Q5.** Explain, with an example, why solving $P'(q) = 0$ alone does not guarantee maximum profit.

Solution

1. $P'(q) = 0$ only finds a stationary candidate, which could be a maximum, a minimum or neither.
2. Example: $P(q) = q^3 - 6q^2 + 9q$ on $[0,5]$ has $P'(q) = 3(q - 1)(q - 3)$. At $q = 3$ there is a local minimum, $P(3) = 0$.
3. Even $q = 1$, a local maximum with $P(1) = 4$, is not the absolute maximum, because the endpoint gives $P(5) = 20$. So test the candidates and compare the endpoints.

Final Assessment · Unit Test

11 questions · 50 marks · K10 / A15 / T15 / C10

Suggested time: 60–75 minutes. Show mathematical working, correct units and written reasoning.

K / 4 **Q1.** For $f(x)=|x|$, determine differentiability at $x=0$ using one-sided derivatives.

Solution

1. f is continuous at $x=0$, because $|x|\to 0=f(0)$ as $x\to 0$.
2. Left derivative: for $h<0$, $[f(h)-f(0)]/h=|h|/h=-h/h=-1$.
3. Right derivative: for $h>0$, $[f(h)-f(0)]/h=|h|/h=h/h=1$.
4. The one-sided derivatives differ, so $f'(0)$ does not exist. f is continuous but not differentiable at $x=0$.

K / 3 **Q2.** State the definitions of velocity and acceleration in terms of position, including SI units.

Solution

1. Velocity is the rate of change of position: $v(t)=s'(t)$, in m/s.
2. Acceleration is the rate of change of velocity: $a(t)=v'(t)=s''(t)$, in m/s^2.
3. Speed is $|v(t)|$. Velocity includes direction and speed does not.

K / 3 **Q3.** State the Extreme Value Theorem, including its hypotheses.

Solution

1. Hypotheses: f is continuous on a closed, bounded interval $[a,b]$.
2. Conclusion: f attains at least one absolute maximum and at least one absolute minimum on $[a,b]$.
3. These values occur at critical numbers or at the endpoints.

A / 5 **Q4.** Given $f(x)=x^2+2$ for $x\le 1$ and $f(x)=mx+b$ for $x>1$, find m and b so that f is differentiable at $x=1$.

Solution

1. Continuity: $f(1)=1^2+2=3$, so $m+b=3$.
2. Left derivative: $2x$ at $x=1$ equals 2. Right derivative: m.
3. Differentiability requires $m=2$.
4. Substitute: $2+b=3$, so $b=1$.
5. Check: the right piece gives $2(1)+1=3$ with slope 2, matching the left piece.

A / 5 **Q5.** For $s(t)=t^2-6t$ (s in metres, t in seconds) over $0\le t\le 5$, find the rest time, displacement and total distance.

Solution

1. $v(t)=2t-6$, so the particle is at rest at $t=3$ s.
2. Positions: $s(0)=0$, $s(3)=-9$, $s(5)=-5$.
3. Displacement $=s(5)-s(0)=-5$ m.
4. Total distance $=|-9-0|+|-5-(-9)|=9+4=13$ m.

A / 5 **Q6.** Find the absolute maximum and minimum of $f(x)=x^3-12x+5$ on $[-3,5]$.

Solution

1. $f'(x)=3x^2-12=3(x-2)(x+2)$, so the critical numbers are $x=-2$ and $x=2$. Both are inside the interval.
2. Evaluate: $f(-3)=14$; $f(-2)=21$; $f(2)=-11$; $f(5)=70$.
3. Absolute maximum $=70$ at the endpoint $x=5$. Absolute minimum $=-11$ at $x=2$.

T / 5 **Q7. Design.** A rectangular poster must contain 384 cm² of printed text, with 6 cm margins at the top and bottom and 4 cm margins on each side. Find the poster dimensions that minimize its total area.

Solution

1. Let the printed region be w by h, in cm, with $wh=384$, so $h=384/w$ and $w>0$.
2. The poster is $(w+8)$ by $(h+12)$, so $A(w)=(w+8)(384/w+12)=480+12w+3072/w$.
3. $A'(w)=12-3072/w^2=0$ gives $w^2=256$, so $w=16$.
4. $A''(w)=6144/w^3>0$, so this is a minimum.
5. $h=384/16=24$. The poster is 24 cm wide and 36 cm tall, with area 864 cm².

T / 5 **Q8. Economics.** A firm faces $p(q)=60-q$ and $C(q)=120+10q$ for $0\le q\le 60$. (a) Find the profit-maximizing quantity and price. (b) A \$10 per-item tax is added to its costs. How do the quantity, price and maximum profit change?

Solution

1. Before the tax: $P(q)=q(60-q)-(120+10q)=50q-q^2-120$.
2. $P'=50-2q=0$ gives $q=25$, and $P''=-2<0$. The price is \$35 and $P(25)=1250-625-120=\$505$.
3. With the tax, cost is $120+20q$, so $P(q)=40q-q^2-120$.
4. $P'=40-2q=0$ gives $q=20$, and $P''=-2<0$. The price is \$40 and $P(20)=800-400-120=\$280$.
5. Output falls by 5 items, the price rises by \$5 and maximum profit falls by \$225. The firm passes only half of the \$10 tax on through its price.

T / 5 **Q9. Biology.** A simplified model for the energy a salmon uses to swim upstream against a 4 km/h current is $E(v)=cv^3/(v-4)$, where $v>4$ is its swimming speed in km/h and $c>0$ is a constant. Find the speed that minimizes energy and state one assumption of the model.

Solution

1. Quotient rule: $E'(v)=c[3v^2(v-4)-v^3]/(v-4)^2=cv^2(2v-12)/(v-4)^2$.
2. $E'=0$ when $2v-12=0$, so $v=6$ km/h. The value $v=0$ is outside the domain.
3. $E'<0$ for $4<v<6$ and $E'>0$ for $v>6$, so $v=6$ is a minimum.
4. Energy tends to infinity as $v\to 4^+$ and as $v\to\infty$, confirming the absolute minimum. The salmon uses the least energy at 6 km/h, which is 1.5 times the speed of the current.
5. One assumption: the current stays constant at 4 km/h for the whole trip.

C / 5 **Q10.** A student writes: “$f'(a)=0$ always means f has a local maximum or minimum at a, and every local maximum happens where $f'(a)=0$.” Correct both claims, giving an example for each.

Solution

1. The first claim is false. $f(x)=(x-1)^3$ has $f'(1)=0$, but $f'(x)=3(x-1)^2\geq 0$ on both sides, so f keeps increasing and has no extremum at $x=1$.
2. The second claim is false. $f(x)=-|x|$ has a local maximum at $x=0$, but $f'(0)$ does not exist.
3. Correct statement: an interior local extremum must occur at a critical number where f is defined ($f'=0$ or f' does not exist). When f is continuous at c and differentiable on either side with a consistent derivative sign on each side, a change from + to − proves a local maximum and − to + proves a local minimum. This is a sufficient test, not a universal necessary condition: a constant function has non-strict local maxima and minima without a derivative sign change. Endpoints are separate candidates for absolute extrema.

C / 5 **Q11.** A driver says, “My acceleration is negative, so I must be slowing down.” Explain in words when this is true and when it is false. Include signs and a numerical example of each case.

Solution

1. Negative acceleration means the velocity is decreasing. On its own it does not say whether the speed is decreasing.
2. True case, $v>0$ and $a<0$: with $v=10$ m/s and $a=-2$ m/s², $v\cdot a=-20<0$, so the car slows down.
3. False case, $v<0$ and $a<0$: with $v=-10$ m/s and $a=-2$ m/s², $v\cdot a=20>0$, so the car speeds up.
4. The driver must compare the signs of v and a, not look at a alone.

ADDITIONAL INTERACTIVE PRACTICE

Instant Feedback Checks

These are extra self-checks. They do not replace the three complete KATC assessments above.

Quick Check 1: Differentiability

A function that is differentiable at x=a must also be:

- ◯ Continuous at x=a
- ◯ Discontinuous at x=a
- ◯ Linear everywhere
- ◯ At a maximum

Answer: Differentiability implies continuity at the same point.

Quick Check 2: Motion

If velocity is −3 m/s and acceleration is −2 m/s², the object is:

- ◯ Moving right and slowing
- ◯ Moving left and speeding up
- ◯ At rest
- ◯ Moving left and slowing down

Answer: Velocity and acceleration are both negative, so speed increases.

Quick Check 3: Optimization

For $P(q)=50q-q^2-120$, the stationary value occurs at:

- ◯ $q=10$
- ◯ $q=50$
- ◯ $q=25$
- ◯ $q=0$

Answer: $P'(q)=50-2q$. Setting it equal to zero gives $q=25$.

THE TEAM

Teacher Profiles

Meet the two student presenters behind this MCV4U Course Culminating Task.

Video Lesson 1

A

Alan
Student Presenter · Motion
MCV4U Grade 12 Calculus & Vectors

Student presenter for differentiability and one-dimensional motion, and the teacher in Video Lesson 1.

Differentiability | Rectilinear Motion | Speed & Distance | KATC Assessment

Theory & Kinematics

Video Lesson 2

J

Jeff
Student Presenter · Optimization
MCV4U Grade 12 Calculus & Vectors

Student presenter for extrema and optimization across design, economics and biology, and the teacher in Video Lesson 2.

Design Optimization | Economics Marginal Profit | Enzyme Activity Model | Drug Concentration Model

Applied Modeling

KEEP LEARNING

Further Learning & References

Free outside resources for each module.

Differentiability

OpenStax Calculus Volume 1, 3.2: The Derivative as a Function (https://openstax.org/books/calculus-volume-1/pages/3-2-the-derivative-as-a-function)

Paul's Online Notes: The Definition of the Derivative (https://tutorial.math.lamar.edu/classes/calci/DefnOfDerivative.aspx)

Velocity & Acceleration

OpenStax Calculus Volume 1, 3.4: Derivatives as Rates of Change (https://openstax.org/books/calculus-volume-1/pages/3-4-derivatives-as-rates-of-change)

Extreme Values

OpenStax Calculus Volume 1, 4.3: Maxima and Minima (https://openstax.org/books/calculus-volume-1/pages/4-3-maxima-and-minima)

Paul's Online Notes: Finding Absolute Extrema (https://tutorial.math.lamar.edu/classes/calci/AbsExtrema.aspx)

Optimization

OpenStax Calculus Volume 1, 4.7: Applied Optimization Problems (https://openstax.org/books/calculus-volume-1/pages/4-7-applied-optimization-problems)

Paul's Online Notes: Optimization (https://tutorial.math.lamar.edu/classes/calci/Optimization.aspx)

Graphing Tool

Desmos Graphing Calculator: try any function from this site (https://www.desmos.com/calculator)

Assignment basis: MCV4U Course Culminating Task Guidelines, Topic C.

Sources: The practice problems were written for this project. The windpipe and salmon models are adapted from classic exercises in university calculus textbooks such as Stewart's *Calculus*.

Models: The enzyme, drug, bird, windpipe and salmon models are simplified classroom models. They are not measured data or medical advice.

Study & Submission Resources

Use the PDF for offline study. All homework and assessment solutions are included.

Download the complete website PDF Requirements & Rubric checklist Printable student response sheet

Original assignment guidelines Original teacher Rubric